

Multiple Choice

1. **Answer:** E

Options: A) Constant energy levels across all trophic levels; B) Increase in nutrient availability at each level; C) Decrease in carbon dioxide concentration; D) Increase in biomass at each successive level; E) Great decrease (90%) in available energy at each level

Note: The correct answer is based on the principle of energy transfer in ecosystems, where approximately 90% of energy is lost as heat at each trophic level, limiting the number of levels that can be sustained in a food chain due to insufficient energy for higher-level consumers.

2. **Answer:** A

Options: A) Sponge; B) Protist; C) Clam; D) Starfish; E) Coral

Note: The correct answer is not a sponge, as sponges are actually classified as part of the benthos rather than the nekton; nekton refers to organisms capable of swimming freely in the water column, such as fish and squid.

3. **Answer:** A

Options: A) an individual's genome; B) a population's genetic variation; C) a species' genetic mutation; D) an individual's genetic makeup; E) a population's gene frequency

Note: Natural selection acts directly on the variations expressed in individuals, which are ultimately determined by their genomes, making an individual's genome the smallest unit upon which natural selection operates.

4. **Answer:** C

Options: A) parenchyma; B) phloem; C) endoderm; D) pericycle; E) cortex

Note: The endoderm is the innermost layer of the root cortex in monocot plants that regulates the movement of water and nutrients into the vascular system, allowing water to enter through the cytoplasm of its cells.

5. **Answer:** A

Options: A) Oncogenes are found only in viruses; B) Oncogenes prevent cancer cells; C) Oncogenes are a type of immune cell; D) Oncogenes detoxify harmful substances in the cell; E) Oncogenes slow down cell division

Note: The correct answer is based on the understanding that oncogenes, which can drive the transformation of normal cells into cancerous cells, are often derived from viral genes, highlighting their association with viral mechanisms of oncogenesis.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The statement is false because environmental factors, such as exposure to carcinogens, infections, and radiation, also play a significant role in the development of cancer alongside inherited genetics and lifestyle factors.

7. **Answer:** True

Options: A) True; B) False

Note: Rotational cleavage involves the orientation of cell divisions occurring at specific angles to the polar axis, particularly in species like mammals, leading to a pattern where the cleavage planes are not strictly vertical or horizontal but rather oblique, which is characteristic of this type of cleavage.

8. **Answer:** True

Options: A) True; B) False

Note: Sound communication in animals is essential for various social interactions and survival strategies, as it enables individuals to attract mates, identify species, defend territories, and assert dominance within their habitats.

9. **Answer:** True

Options: A) True; B) False

Note: The statement is true because a standard deck of cards contains 52 unique cards, and there is only one ace of diamonds, making the probability of drawing it 1 out of 52.

10. **Answer:** False

Options: A) True; B) False

Note: The zona pellucida is a glycoprotein layer surrounding the oocyte and early embryo in mammals, primarily serving to facilitate fertilization and prevent polyspermy, rather than providing protection against desiccation and injury, which is the role of the amniotic fluid and extraembryonic membranes in terrestrial vertebrates.

Long Form Response

11. **Answer:** Animals living in hot arid environments have evolved various adaptive features to enhance their survival, one of which includes the structure of their kidneys, particularly the loops of Henle. In these animals, the loops of Henle tend to be shorter, which facilitates the efficient reabsorption of water during urine formation. This adaptation allows for maximal water conservation, crucial for maintaining hydration in environments where water is scarce. By producing more concentrated urine, these animals can reduce water loss and survive prolonged periods without drinking. Thus, the morphology of the loops of Henle plays a vital role in their ability to thrive in challenging conditions.

Note: correct because it accurately describes how the structure of the loops of Henle in

the kidneys of animals in hot arid environments is adapted to enhance water reabsorption and produce concentrated urine, thereby facilitating effective water conservation essential for survival in such conditions.

12. **Answer:** Swallowing involves a coordinated sequence of physiological mechanisms to ensure that food and air are directed into their respective pathways. When food or liquid is ingested, pressure receptors in the pharynx are stimulated, triggering a reflex that elevates the larynx. This elevation causes the epiglottis to fold down over the glottis, effectively sealing off the airway and preventing aspiration. As a result, the bolus of food is channeled into the esophagus, which is situated behind the trachea. This intricate interplay between the pharynx, larynx, and epiglottis is essential for safe swallowing, allowing simultaneous respiratory and digestive functions to occur without interference.

Note: correct because it accurately describes the reflexive actions of the pharynx, larynx, and epiglottis during swallowing, highlighting how the elevation of the larynx and the closure of the epiglottis prevent food from entering the airway and ensure proper routing into the esophagus.